# KO-Shadow: KnOwledge-Driven Shadow Progressive Removal Framework for Very High Spatial Resolution Remote Sensing Imagery

Yang Yang, Mingqiang Guo, Qiqi Zhu, *Senior Member, IEEE*, Longli Ran, Jun Pan, and Jiancheng Luo

***Abstract*— The formation of shadows in very high spatial resolution (VHR) remote sensing imagery is attributed to light being blocked by objects, reducing spectral radiance in the shadow landscape. An accurate and robust shadow removal method can recover spectral and textural information and, hence, is a crucial preprocessing step for urban image analyses. In this study, we develop a KnOwledge-driven shadow progressive removal (KO-Shadow) framework with three subnets for VHR imagery using a weakly supervised manner. Specifically, the shadow preelimination subnet is proposed to initially address the large chromatic aberration between the real and shadow situations. Then, the prior knowledge-guided refinement subnet is proposed to refine the preelimination results by mining tone and texture information. Moreover, the locality feature discriminator is designed for region-specific evaluation of the generated shadow-free samples to improve the capacity of subnets. Experimental results of six typical cities in the world show that KO-Shadow is superior to the existing methods. Moreover, the generalizability analysis in complex urban scenarios validates the robustness of our method. The shadow recovery score (SRI) is proposed to evaluate the spectral similarities between the recovered area and shadow-related land-cover types (e.g., road, building, and lawn). The results show that KO-Shadow can yield more visually realistic shadow-free images and better quantitative performance. Overall, KO-Shadow provides a new perspective for VHR image shadow removal by mining the prior knowledge of the complex shadows in urban areas.**

***Index Terms*— Knowledge-driven, progressive refinement, shadow removal, very high spatial resolution (VHR), weakly supervised.**

## I. Introduction

SHADOWS are generally found in very high spatial resolution (VHR) remote sensing imagery, which is formed of complex physical interactions between light sources (e.g., the Sun, surface, and properties of the geographical objects in images) [1]. By providing outdoor thermal comfort, shadows can alleviate the urban heat island effect and are deemed to be the key to building identification [2], [3]. However, shadows generally lead to significant spectral gaps between the sunlit land-cover and shadow counterparts, which causes a challenge for accurate land-cover classification [4]. In addition, due to the reduced light intensity in the target tracking tasks, the object will be lost when it enters the shadow area [5]. To eliminate the reduction of radiometric information in shadow areas and improve the quality of VHR imagery, it is necessary to realize shadow removal tasks in urban areas [6], [7]. However, the absence of paired shadow-free labels, as well as the complex spectral and texture that appeared in VHR images in urban areas, presents great challenges for shadow removal.

Over the past years, many researchers have tried to resolve spectral inconsistencies caused by shadows. According to different model complexity, these methods can be mainly divided into three categories [8], [9], [10]: 1) mathematical methods; 2) machine learning methods; and 3) deep learning methods. Traditional mathematical methods remove shadows by constructing the mathematical relationship between the paired shadow and shadow-free areas. Compared to the traditional methods, machine learning requires manually dividing specific samples [22] to form a shadow and nonshadow library, which are correlated by the Markov random field (MRF). In addition, due to the rich spectral information and multiscale geometric features of the VHR images, the aforementioned methods of stacking multiple steps easily lead to the accumulation of errors and ultimately affect the results [23], [24], [25], [26].

In recent years, deep learning has employed a fully automated feature simulation framework to learn basic information [27], [28], [29], [52], [53], [54], [55], demonstrating superior intrinsic feature extraction capabilities in an end-to-end approach [30], [31]. There are some shadow removal methods proposed for natural shadow imagery, which can be mainly divided into supervised shadow removal, weakly supervised, and unsupervised shadow removal methods. Supervised methods require a large number of paired shadow and shadow-free samples for the model training, which is unrealistic for VHR images [32], [33], [34]. Although weakly supervised methods obviate the need for paired datasets, they still require nonshadow images from the same region to learn textural and spectral information. This requirement presents a significant challenge for shadow detection in VHR imagery [35], [36]. Unsupervised methods utilize pretrained

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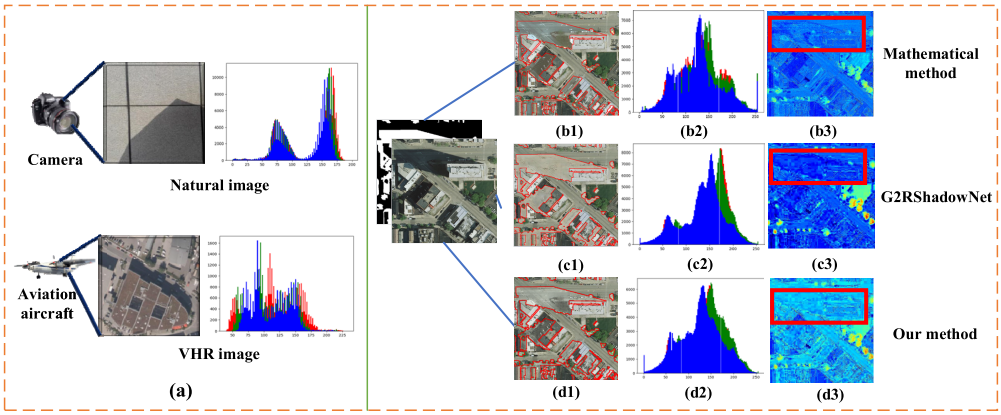


Fig. 1. (a) Histogram distribution of different channels for natural and VHR images. (b1) Shadow removal result of mathematical method. (b2) Color histogram obtained from (b1). (b3) Feature map extracted by pretrained VGG-16 from (b1). (c1) Shadow removal result of computer vision method, i.e., G2RShadowNet. (c2) Color histogram obtained from (c1). (c3) Feature map by pretrained VGG-16 from (c1). (d1) Shadow removal result of our method. (d2) Color histogram obtained from (d1). (d3) Feature map by pretrained VGG-16 from (d1).

models to generate shadow-free images, thereby addressing the issue of data scarcity. However, these methods often fail to effectively remove shadows from VHR images, particularly in the diverse urban landscapes around the world.

The current issues associated with VHR image shadow removal can be summarized as follows.

1) Classic shadow removal methods require carefully defining features over each class on individual scenes to improve optimization capability. Due to the large spectral variability of the VHR imagery in urban areas, varying types of shadows, and sophisticated spatial details, a data-driven end-to-end framework is desired to address the limitations of traditional methods. However, the insufficient paired data led to a nuisance for the data-driven mechanism, and previous shadow removal was performed using a single network, where the effectiveness of the model was directly determined by how well it was trained, which greatly limits shadow removal effects.
2) As shown in Fig. 1(a), the VHR shadow area includes diverse textural details and spatial configuration, which caused the histogram distribution of individual channels to be more complex than natural images. As can be seen in Fig. 1(b1) and (d1), the variety of textures will cause significant spectral contrast in recovered areas by the mathematical method, and the histogram distribution [see Fig. 1(b2)] is more chaotic compared with our method [see Fig. 1(d2)]. Nevertheless, as shown in Fig. 1(c1) and (d1), the internal texture information will be lost using the computer vision method (G2RShadowNet), and the recovered categories in the feature map are more ambiguous compared to the mathematical method, as shown in the red boxes in Fig. 1(b3) and (d3).
3) For the VHR imagery shadow removal task in complex urban environments, the shadows and nonshadows exist in the same image, making it difficult to obtain pure shadow-free images for discriminator correction. The traditional discriminator of the generative adversarial network (GAN) takes the whole image as input when discriminating the reality of the generated shadow-free images. This makes it hard to circumvent the interference of shadow regions for the discriminators, as well as weakening the efficiency of the shadow-free generator.

Considering the characteristics of irregular urban shadows in VHR images, a KnOwledge-driven shadow progressive removal framework (KO-Shadow) is proposed to effectively eliminate shadows in urban areas. Specifically, a shadow preelimination subnet constructed through the interaction of the shadow generator, shadow-free generator, and locality feature discriminator is designed to remove shadows, narrowing the visual discrepancy initially. The prior knowledge-guided refinement subnet further removes shadows by referring to the angle prior to direct sunlight in each image, as well as the prior spectral and textural knowledge, combining nonshadow regions and simulated pseudoshadow regions. The locality feature discriminator updated by the pure shadow or nonshadow features is designed to effectively discriminate the authenticity of the generated shadow-free images.

The main contributions of our work are given as follows.

1) A shadow removal framework (KO-Shadow) is proposed to address the difficulties of VHR image shadow removal. KO-Shadow only requires the original VHR image and the shadow mask to achieve model training and shadow removal. Compared with the traditional methods, KO-Shadow performs stronger robustness and can generate the most realistic spectral and texture information in a weakly supervised learning manner. In contrast to previous deep learning models, the network architecture is constructed through incremental training of three subnets, which simplifies model fitting and overcomes the limitation of a single network.
2) To alleviate the significant spectral and texture difference between the recovered and real nonshadow regions, the prior knowledge-guided refinement subnet is designed to mine the prior knowledge of the complex shadows in urban areas. The shadow spectrum recovery is guided by the specific area with similar color distribution, which is obtained based on the angle prior of direct

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sunlight. In addition, the pseudoshadow region is simulated to provide prior texture knowledge for textural feature recovery in shadow areas. Based on the prior knowledge-guided refinement subnet, KO-Shadow can progressively recover realistic feature distribution.

3) To further improve the performance of the weakly supervised learning framework, a locality feature discriminator is proposed to overlook the useless shadow disturbance. Different from previous methods conducting the discrimination from the whole image, this module only performs partial convolution operations on the generated shadow-free regions with the constraint of the shadow mask. This improves the discrimination capacity and promotes the performance of the generator. Through this novel generator–discriminator cooperative scheme, the newly generated images from the shadow preeliminator can produce more realistic urban areas.

The remainder of this article is organized as follows. Section II specifies related work. Section III introduces our proposed KO-Shadow framework for VHR imagery shadow removal. The detailed validation and assessment of experimental results are discussed in Section IV. Finally, the conclusions and future outlook are presented in Section V.

## II. Related Work

Traditional methods mainly include linear correction, histogram matching correction (HMC), gamma function, and gradient model. For linear correction, the shadow is modeled as a combination of additive and multiplicative noise. Then, the nonshadow pixels to the first order are recovered by a linear function [11], [12]. HMC methods apply statistical correction to match the gray-scale distribution of the associated regions, which can be used to enhance the local comparison without affecting the overall consistency [13], [14]. The gamma function considers shadows as a multiplicative noise destroying the luminance of the underlying pixels. This method manually determines the gamma value and employs a power function to establish the relationship between shadow and nonshadow pixels [15], [16], [17], [18]. The gradient model recovers the shadow by eliminating the luminance gradient and shadow boundary [19], [20], [21].

To address the model complexity of traditional methods, deep learning-based shadow removal methods have been proposed, which are mainly divided into supervised, weakly supervised, and unsupervised methods. Owing to the availability of paired natural image shadow datasets, a series of supervised methods can be trained on images from the same region with and without shadows. Le and Samaras [34] combined deep learning with linear correction to obtain the corresponding model parameters from paired shadow datasets, and then, the shadow area is corrected according to the trained linear function. Hu et al. [33] first introduced attention weights in a spatial recurrent neural network (RNN) to form a direction-aware attention mechanism when aggregating spatial contextual features. He et al. [32] designed a well-designed masked adaptive instance normalization (MAdaIN) mechanism with embedded aligners to produce hidden features that consider an illumination consistency of different regions. In addition, it can also separate deal feature statistics for shadow and shadow-free areas based on the shadow mask. However, although the supervised method can remove shadows better according to the labels, there are still chromatic aberrations in the matched dataset due to the different capture times, which influenced the ability of the model. Besides, the lack of paired datasets for the VHR imagery shadow removal task greatly limits the development of supervised methods. To solve the problem of the difficult availability of paired datasets, the weakly supervised methods use unpaired datasets for training, which only require shadow and shadow-free data in the same environment. Hu et al. [35] used adversarial training by constructing two generators, one generator is used to remove shadows, and the other generator generates shadows from the shadow-free image and the mask. Liu et al. [36] initially compensated the luminance to train a CNN module and then trained a new CNN module to evaluate the final shadow removal under the guidance of the luminance information. Although the weakly supervised methods do not require a paired dataset, they still require shadow-free images, which is also difficult to apply to VHR images.

In remote sensing, due to the difficulty in obtaining paired data, few deep learning methods are proposed for shadow removal. Luo et al. [51] constructed a paired shadow dataset, named the UAV-SC dataset, by using a UAV to take photographs of the ground surface at different moments on the same day. Based on this paired dataset, a GAN network is constructed for shadow removal. However, the color inconsistency between this paired dataset exists, which is caused by the acquisition time differences between shadow and nonshadow images (e.g., shadow images are taken at noon, and nonshadow images are taken in the morning or afternoon). In addition, it usually has limited transferability for networks trained on UAV image pairs to satellite pairs, and nonshadow satellite images are still inaccessible. Because of the high costliness of accessing matched nonshadow satellite data, it remains urgent for a weakly supervised shadow removal method without nonshadow images.

## III. Methodology

The proposed KO-Shadow framework includes four main parts, as shown in Fig. 2.

1) *Shadow Preelimination Subnet:* To generate a realistic shadow-free image, the shadow preelimination subnet was trained with the constraints of identity consistency and cycle consistency. The shadow-free generator is able to initially narrow the feature gap, simplify the complexity of model fitting, and then provide support for subsequent processing.
2) *Prior Knowledge-Guided Refinement Subnet:* The prior knowledge base is constructed, including the spectral prior in nonshadow regions and textural prior in simulated pseudoimages. Based on the sunlit angle-guided shadow-free generator and the guidance of prior knowledge, the shadow-refinement generator is trained to further recover the spatial and chromatic information.

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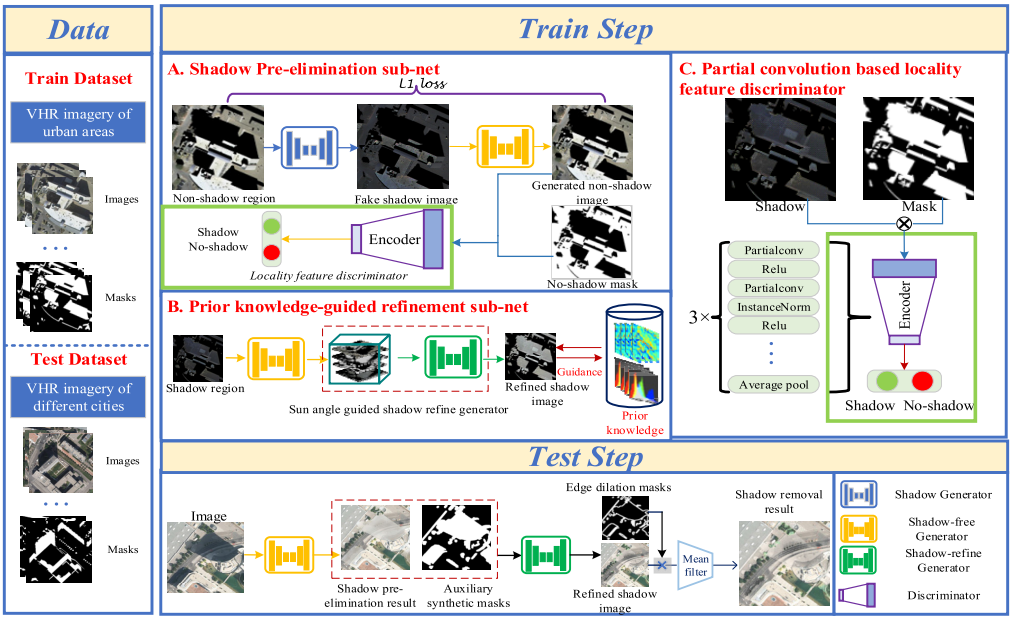


Fig. 2. Shadow removal based on the KO-Shadow framework for VHR images.

3) *Locality Feature Discriminator:* The locality feature discriminator is designed to discriminate the reality of generated features for specific locations based on the input mask. In this way, this discriminator can reduce the interference of the remaining shadow areas and further enhance the performance of the shadow preeliminator.
4) *Test Phase:* The original VHR image is first input into the trained shadow-free generator to initially reduce the shadow phenomenon. Then, the refined nonshadow image is acquired based on the sunlit angle prior-guided shadow-refine generator. The spatial and chromatic information were further recovered by the guidance of feature prior information from the nonshadow regions. To address the effect of edge misalignment between the manual annotation masks and shadows, median filtering was utilized for the mitigation of edge effects in the test phase, which generated edge buffers based on the solar incidence angle. We first extract the edges of the shadows and then directionally inflate the edges according to the angle of the Sun, thus avoiding smoothing the edges of the building. In this way, we can effectively alleviate the smoothing of the building boundary areas. We have tried three filters, including Gaussian filtering, median filtering, and mean filtering; both Gaussian and median filters produce noticeable edge effects, and the median can alleviate boundary effects, so we finally chose the median filter for shadow edge smoothing.

### A. Shadow Preelimination Subnet

Due to the high complexity of VHR images in urban areas, the direct use of a single model to remove shadows will cause the result to be difficult to be realistic. Therefore, the shadow preelimination subnetwork is first constructed to perform the preliminary removal and simplify the whole network. This subnet scarcely performs any texture and color supervision and only uses global consistency, thus providing the basis for further shadow removal.

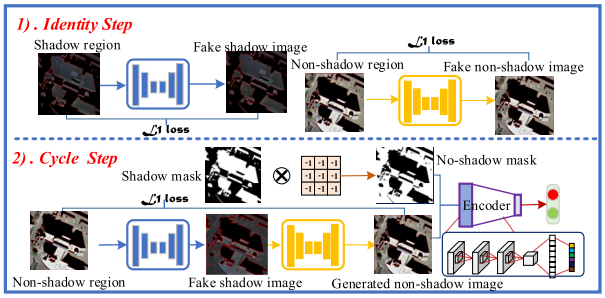


Fig. 3. Shadow preelimination subnet framework.

Although it is possible to obtain a small area of shadow-free images by cropping, the low resolution of the cropped shadow-free images and the small number of feature categories still hardly satisfy the demand for removing VHR image shadows. Therefore, as shown in Fig. 3 of the shadow region and non-shadow region, two types of "pure" data were simulated to initialize the training generator.

As shown in Fig. 3, two generators ($G_s$ and $G_f$) motivated by [35] were employed to initially generate and remove shadow areas. However, the network proposed by Hu et al. [35] requires unpaired shadow and nonshadow images for training. Due to the lack of no-shadow VHR images, we converted the model to accept the synthesized part of shadow or no-shadow images for training.

First, a convolution layer of kernel size $7 \times 7$ is used to expand the receptive field, which captures the larger structural and textural information in the image. Then, an instance normalization layer and an ReLU activation layer are followed. In this way, the shallow features are extracted. Then, two convolution layers of kernel size $3 \times 3$ with stride $2 \times 2$ are applied to reduce feature size. After that, the features are filtered by nine residual blocks. The deconvolution layer progressively enlarged the feature resolution and adopted instance normalization and ReLU after each convolution operation to transform the features to between $-1$ and 1. In this way, the same resolution images are generated by the shadow generator.

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This stage consists of three main progressive ways to remove shadows. First, the shadow and nonshadow regions were fed into two generators ($G_s$, $G_f$), respectively, to generate new shadows and nonshadows. In this way, the input and output of the two generators should be consistent. With this principle, two generators were simply initialized using identity consistency, as shown in (1) and (2). To further train the ability of the generator, the cycle consistency principle was adopted to constrain the network, as shown in (3). In detail, since the initialized generator has simple shadow generation and removal capabilities, a fake shadow image was generated using $G_s$ for the nonshadow image. Subsequently, a fake nonshadow image was generated using $G_f$. In this case, the image after shadow removal should be consistent with the image of the input $G_s$, whose constraint could be used to further train generators

$$\mathrm{Sd}' = G_s(\mathrm{Sd}) \tag{1}$$

$$\mathrm{Nsd}' = G_f(\mathrm{Nsd}) \tag{2}$$

$$\mathrm{Nsd}'' = G_f(G_s(\mathrm{Nsd})) \tag{3}$$

$$D(\text{fake-sd}) = 0//1. \tag{4}$$

When the generator has finished training, the evaluation of the discriminator can better guide the generator training. The locality feature discriminator was employed to determine the reality of the generated image, which effectively drives the generator to generate a more realistic image. As shown in (4), if the generated sample is a shadow, it is closer to 1. On the contrary, the value is more relative to 0. Subsequently, the generated fake shadow image with the corresponding mask was combined to make a convolution process on the mask-covered area, judging the differences with the real values. After completing the whole training, a hybrid loss function was used for the supervision of the model so that the backward supervision could be performed correctly and efficiently.

Here, $G_s$ represents the shadow generator, $G_f$ represents the shadow-free generator, Sd represents the shadow region, Sd′ represents the generated false shadow image, Nsd represents the nonshadow region, and Nsd′ represents the generated false no-shadow images. Nsd″ represents the generated nonshadow image. $D$ denotes the locality feature discriminator, and fake-sd denotes the fake shadow image that was generated by $G_s$. The different loss functions were used for each phase to constrain the training of the model. For identity consistency, the $L_1$ loss is used to discriminate the loss for generating shadows and generating nonshadows. The adversarial loss contains discriminator loss and cycle consistency loss, where mse loss is used for discriminator loss and $L_1$ loss for cycle consistency loss. The GAN loss function form is shown as follows:

$$L_{\mathrm{GAN}} = L_{\mathrm{Cycle}} \times 10 + L_{\mathrm{Dis}}. \tag{5}$$

The final loss function is obtained by weighting the individual component losses

$$L = L_{\mathrm{Iden}}(G_s) \times 5 + L_{\mathrm{Iden}}(G_f) \times 5 + L_{\mathrm{GAN}}. \tag{6}$$

### B. Prior Knowledge-Guided Refinement Subnet

*1) Prior Knowledge-Guided Refinement Framework:* There is still a significant discrepancy with the real situation after preliminary recovery. By analyzing the shadow characteristics in remotely sensed images, it is found that shadow removal is particularly important in both spectral and textural aspects. Therefore, a prior knowledge-guided refinement subnet was proposed to further improve the effect of shadow preelimination results. The structure is similar to the shadow preelimination subnet, the main difference being that the input is learned by overlaying the image and the shadow mask. As shown in Fig. 4, the subnet is dedicated to restoring the texture and color information of the shadow area, respectively, by two synchronous operations. One part is used to ensure that the spectral of the shadow areas in the same image is consistent with the nonshadow areas. The other part is maintaining the texture coverage category before and after elimination. In addition, according to the first law of geography, the closer the things are related, the stronger the correlation [38]. Thus, outer shadow edge information is essential in recovering shadows. However, the VHR shadow is caused by the reduction of light in a particular direction due to the blockage of tall buildings; only the part of the nonshadow area extended along the angle of solar incidence has a complementary intellectual effect.

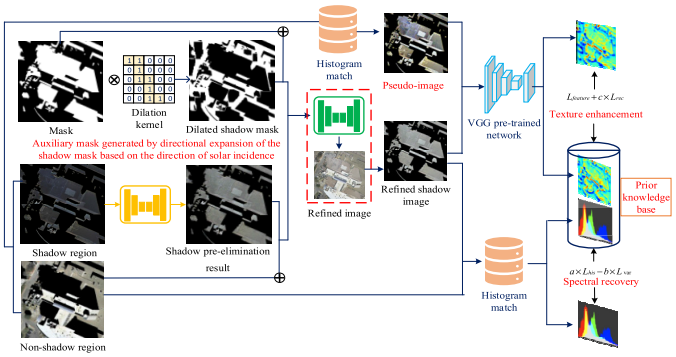


Fig. 4. Prior knowledge-guided refinement subnet framework.

Thus, the shadow elimination auxiliary information area was formed according to the angle prior of direct sunlight after data preprocessing, and the final data are shown in Fig. 5.

The input to the optimizer is a four-channel tensor consisting of a three-channel image and a one-channel mask. In detail, Fig. 5(a) and (b) were synthesized into a new three-channel image and Fig. 5(d) and (e) into a one-channel mask tensor, which is fed into Gr to produce the refined result.

*2) Spectral Recovery Based on Histogram Loss:* The spectral distribution of the shadow and nonshadow areas should be consistent in the same image. Hence, we modified the color migration [39], which initially only matches the spectral distribution of different images with histogram loss, to perform spectral matching of different areas in the same image. First, according to the mask, the shadow and nonshadow regions were extracted, respectively, from the refined shadow image and the original image [as shown in (7) and (8)]. Transform each dimension of the two areas into a 2-bit spatial vector based on the relationship between the RGB values

$$\mathrm{Sd}_{i,j} = \{x_{i,j} | \mathrm{Mask}_{xi,j=1}\} \tag{7}$$

$$\mathrm{Nsd}_{i,j} = \{x_{i,j} | \mathrm{Mask}_{xi,j=0}\}. \tag{8}$$

As shown in (9) and (10), taking the shadow region $R$ channel as an example, where $\in$ is a minimal value, which

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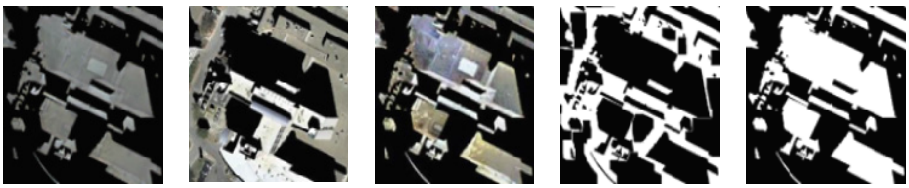


(a) (b) (c) (d) (e)

Fig. 5. Data required for the prior knowledge-guided refinement subnet. (a) Fake no-shadow images after initial shadow removal by $G_f$. (b) Nonshadow regions. (c) Simulated shadow region after histogram matching. (d) Sunlit angle-guided dilated shadow masks. (e) Shadow mask.

is used to prevent the fraction of 0, and $x$ is the pixel value belonging to the shadow region

$$IvR(x \in \mathrm{Sd}_{i,j}) = \log\left(\frac{IR(x \in \mathrm{Sd}_{i,j}) + \in}{IG(x \in \mathrm{Sd}_{i,j}) + \in}\right) \tag{9}$$

$$IuR(x \in \mathrm{Sd}_{i,j})) = \log\left(\frac{IR(x \in \mathrm{Sd}_{i,j}) + \in}{IB(x \in \mathrm{Sd}_{i,j}) + \in}\right). \tag{10}$$

Through the above calculation, a 2-D $vu$ coordinate was summarized in every channel. For a specific region, three different sets of $uv$ coordinates could be formed to describe the color property characteristics of the area. In addition, the inverse-quadratic kernel $k(Iuc, Ivc, u, v)$ was calculated to map the samples to a higher dimensional linearly divisible space, which is defined as

$$k(Iuc, Ivc, u, v) = \left(1 + \frac{|Iuc - u|^2}{\tau}\right)^{-1} \times \left(1 + \frac{|Ivc - v|^2}{\tau}\right)^{-1} \tag{11}$$

where $c \in \{R, G, B\}$ and $\tau$ is a decay parameter to control the smoothness of the histogram and make it differentiable. Due to the variation in the overall distribution of each shadow pixel, $Iy(x)$ is calculated to describe the contribution of each pixel, which is defined as $Iy(x) = (I_R(x)^2 + I_G(x)^2 + I_B(x)^2)^{1/2}$. The final unnormalized histogram is calculated as

$$H(u, v, c) \propto \sum_x k(Iuc(x), Ivc(x), u, v) \times Iy(x). \tag{12}$$

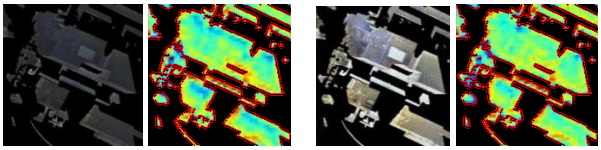
The $H$-vectors of the shadow and nonshadow regions were extracted in the above way. Finally, they transformed into 64 × 64 × 3 feature vectors, which ensures the consistency of the colors of the two areas by approximating the color features. As shown in (13), $H_{\mathrm{nsd}}$ represents the histogram feature of the nonshadow region, $H_{\mathrm{sd}}$ represents the histogram feature of the refined shadow area, and $\|$ represents the $L_1$ loss function

$$L_{\mathrm{his}} = \frac{1}{\sqrt{2}} \| \sqrt{H_{\mathrm{nsd}}} - \sqrt{H_{\mathrm{sd}}} \|. \tag{13}$$

Histogram loss is difficult to ensure global color uniformity due to the use of transformed feature maps for color matching. To alleviate these consequences, a variance loss is added to increase the variance difference of the smoothed colors in each image, which can be described as

$$L_{\mathrm{var}}(I_{\mathrm{sd}}, I_{\mathrm{nsd}}) = -w \sum_{c \in \{R,G,B\}} |\sigma(I_{\mathrm{nsdc}} * G) - \sigma(I_{\mathrm{sdc}} * G)| \tag{14}$$

where $w = \|H_{\mathrm{nsd}} - H_{\mathrm{sd}}\|$ is a weighting factor, which is used to calculate the contribution of each pixel. Due to the histogram $H_{\mathrm{nsd}}$ of the non-shadow regionand the histogram $H_{\mathrm{sd}}$ of the refined shadow areahave large difference, $\delta$ is used to mitigate the standard deviation of its input by using a Gaussian blur kernel with a kernel of 15. As shown in (14), Insd represents the input of the nonshadow region, $I_{\mathrm{sd}}$ represents the input of the shadow region. $I_{\mathrm{nsdc}}$, and $I_{\mathrm{sdc}}$ indicate the $I_{\mathrm{nsd}}$ and $I_{\mathrm{sd}}$ on each channel, respectively.



(a) (b)

Fig. 6. (a) VGG feature map of the shadow region. (b) VGG feature map of the simulated pseudoshadow region.

*3) Texture Enhancement Based on Feature Loss:* Although many current methods can recover color information, the recovered texture of the shadows will be lost and appear blurred. To solve this problem, a combination of deep learning feature extraction networks and prior knowledge was adopted to ensure the integrity of the interior features. Furthermore, the simulated pseudoimage can also provide spatial knowledge for feature reconstruction.

Many shadow removal methods are available to provide simulated pseudoimages, such as linear correction, gamma function, and machine learning. However, compared with other methods, the histogram matching method is simplified. Its contrast of feature categories is obvious after the recovery, which can also better facilitate the extraction of features. For feature extraction networks, AlexNet [40], ResNet [41], ReXsNet [42], and so on show excellent information extraction ability, but the VHR images have high category complexity and large scale, which undoubtedly increases the complexity of the feature extraction. In general, VGG19 has a more straightforward structure and deeper layers [43].

As shown in Fig. 5(c), there is a significant chromatic aberration in the image after histogram deshadowing, but the feature type is more obvious than the original image. The VGG pretraining model extracts features from the refined shadow images and histogram equalized images. As shown in Fig. 6, a 20th layer is more robust to shadow through several experiments, which means a minimal difference in features extracted from refined shadow and pseudoimages. Finally, the $L_1$ loss is used to make the optimized refined shadow features approximate the features of the pseudoimage as closely as possible, which is defined as (15), where $\|$ represents the $L_1$ loss function and VGG() reparents features extracted by the VGG model

$$L_{\mathrm{feat}} = \|\mathrm{VGG}(\mathrm{ref_sd}) - \mathrm{VGG}(\mathrm{fake_nsd})\|. \tag{15}$$

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Although feature loss can well keep the overall texture information, the detail information is challenging to save. Therefore, the Laplace operator is employed to extract the image edge information and use reconstruction loss to ensure the detailed information better

$$L_{\text{rec}}(I_{\text{ref_sd}}, I_{\text{fake_nsd}}) = \|I_{\text{ref_sd}} * L - I_{\text{fake_nsd}} * L\| \tag{16}$$

where $^*L$ represents the Laplace operator, which can not only suppress the image color information but also better extract the edge detail features in the image. The reconstruction loss was employed to maintain detailed features with the VGG extraction, which can effectively ensure the maintenance of feature consistency before and after shadow removal.

The loss function of the final optimizer is obtained by weighing the different losses. However, an excessive proportion of color loss will lead to an overall convergence of the recovered area, resulting in the loss of texture information. In contrast, when the feature loss is too significant, it will lead to an obvious color difference between the recovered area and the real shade-free area, and the consistency of the image cannot be guaranteed. Through several experiments, we have determined the loss function weights that give the best results, which are defined as follows, where $a$ is 0.7, $b$ is 0.15, and $c$ is 1.5:

$$L_{\text{ref}} = a \times L_{\text{his}} - b \times L_{\text{var}} + L_{\text{feature}} + c \times L_{\text{rec}}. \tag{17}$$

### C. Partial Convolution-Based Locality Feature Discriminator

The original discriminator discriminates the entire urban area distribution as one unit, which makes it confusing for the shadow and nonshadow areas and leads to discriminator training failure. As shown in Fig. 5(a), the discriminator capability will be interfered with by the useless distribution because of the filling of all nonshadow areas, which further affects the shadow preeliminator quality. To address this issue, the locality feature discriminator was proposed using partial convolution [44] instead of the original convolution to determine regions obtained by the original image and corresponding shadow mask. The partial convolution is defined as follows:

$$x = \begin{cases} W^T \times \left( X(i,j) \times M(i,j) \times \dfrac{1}{\|M\|} \right) + b, & \|M\| > 0 \\ 0, & \|M\| = 0. \end{cases} \tag{18}$$

As shown in (18), partial convolution also uses the same combination of linear functions as standard convolution, which consists of weights $W$ and deviations $b$. $X$ is the pixel value in the currently defined convolution kernel, while $M$ is the corresponding mask. In this way, according to the mask, only its labeled position is convolved during convolution, which avoids the disturbance of other areas. Furthermore, to ensure that the feature map is after each convolution on the generated feature, the mask is also updated after each forward propagation convolution, which is defined as follows:

$$m = \begin{cases} 1, & \|M(i,j)\| > 0 \\ 0, & \|M(i,j)\| = 0. \end{cases} \tag{19}$$

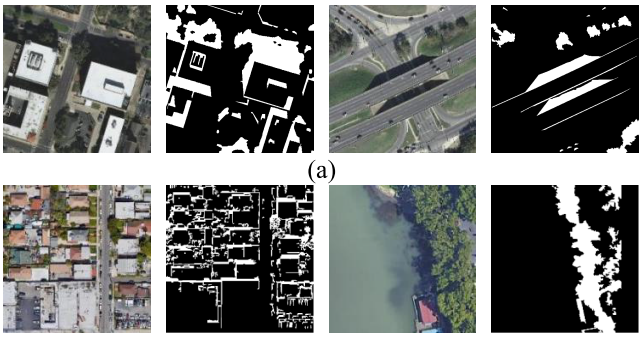

Fig. 7. (a) Image samples in the AISD dataset. (b) Image samples in the Rome dataset.

The effective region features are extracted by four $4 \times 4$ partial convolutions with stride $2 \times 2$, obtaining the preprocessed shadow features. The extracted shadow features are compressed to one channel by a $4 \times 4$ partial convolution to extract the most significant features. Finally, the compressed values are pooled on average to further verify the authenticity of the generated images. Note that the instance normalization and ReLU were performed in each partial convolution to normalize the data. To ensure the accuracy of the discriminator, the mse loss function is used to update the discriminator at the end of each iteration based on the true shadow area and the shadow-free image

$$L\text{Dis}(\text{real}) = \|D(\text{sd}, \text{sd_mask}) - 1\| \tag{20}$$

$$L\text{Dis}(\text{fake}) = \|D(\text{Nsd}, \text{Nsd_mask}) - 0\|. \tag{21}$$

The final discriminator loss function is

$$L\text{Dis} = (L\text{Dis}(\text{real}) + L\text{Dis}(\text{fake})) \times 0.5. \tag{22}$$

## IV. Experiments and Analysis

### A. Study Area and Data

*1) AISD Dataset:* The AISD dataset (https://GitHub.com/RSrscoder/AISD) is a VHR shadow detection dataset covering five cities in the world [37]. Compared to other datasets, the AISD dataset was widely used because of its high quality and wide coverage. In addition, the AISD data are rich in shadow categories and also contain vegetation shadows, such as trees, which increases the difficulty of shadow detection and removal. Furthermore, the AISD training dataset consists of 412 images and corresponding shadow masks. Because of the different image sizes in this dataset, the AISD dataset was carefully constructed (including data division, label revision, and so on) to standardize the resolution and perform data enhancement. Finally, the AISD dataset is constructed of 2135 images with a size of $400 \times 400$ by randomly cropping with a step size of 128.

*2) Rome Dataset:* To verify the robustness of the proposed framework in more complex spatial situations, the Rome dataset was constructed using 0.3-m images from Rome, which were obtained from the Google Earth platform for the training and validation of the proposed model. As shown in Fig. 7, compared with the AISD dataset, the shadow has finer shaded and more sophisticated shapes. Finally, the Rome dataset has

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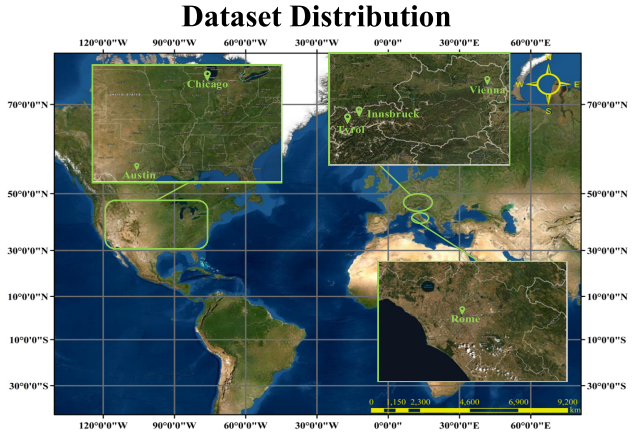


Fig. 8. Overview of the dataset distribution. The images were collected from Austin, Chicago, Vienna, Innsbruck, Tyrol, and Rome.

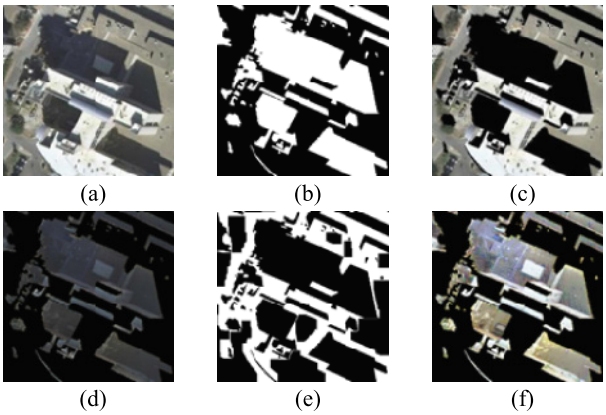


Fig. 9. (a) Original images. (b) Shadow masks. (c) Nonshadow regions. (d) Shadow regions. (e) Angle prior of direct sunlight dilated shadow masks. (f) Simulated pseudo-no-shadow regions formed based on (d).

a total of 231 image patches with a size of 512 × 512, which contains abundant shadows. Each image with its shadows was labeled following a comprehensive annotation pipeline. We followed the same data processing as the AISD dataset, which cropped the images to 400 × 400 according to a step size of 128. Finally, the Rome dataset containing 918 images was obtained.

*3) Complex Scenario Mapping:* To evaluate the generalizability of the proposed method, a complex scenario mapping is constructed, which is a manually labeled pixel-level VHR urban imagery covering Rome. The shadow region is labeled by image interpretation, which has a size of 4096 × 2048 pixels and a resolution of 0.3 m/pixel. The validation of the complex scenario mapping further confirms the superiority of our method, especially the robustness of the KO-Shadow to different imaging conditions and Chaotic scenes. The distribution of the overall data is shown in Fig. 8, covering Austin, Chicago, Vienna, Innsbruck, Tyrol, and Rome.

### B. Data Preprocessing

For the VHR image shadow removal, the acquisition of shadow-free images is usually inaccessible. To solve the problem of absent data, four classes of auxiliary images were synthesized based on the original images [see Fig. 9(a)] and the corresponding shadow mask [see Fig. 9(b)]. As shown in Fig. 9(d), the shadow region was constructed based on the shadow mask, which was produced by retaining the RGB values for the shadow part and deleting the pixel values for the rest of the nonshadow part. Similarly, the nonshadow region was generated by using the same approach [see Fig. 9(c)]. Due to the presence of shadows, feature categories located inside the shadow areas are difficult to mine, owing to the low contrast. To enhance the detailed information inside the shadows in the prior knowledge-guided refinement subnet, the simulated pseudoshadow region [see Fig. 9(f)] is synthesized using the shadow region [see Fig. 9(d)] and nonshadow region [see Fig. 9(c)] for histogram matching to intensify texture features. In addition, for VHR urban image shadows, the major formation reason is that buildings block sunlight, and only the side along the sunlight angle has a similar texture and spectral information with the shadow region. There are five cities in the AISD dataset and the same sunlight angle in the same city, so we have experimented with the expansion angle and coefficients of the city shadow several times and, finally, determined the direction of the expansion kernel and the kernel size of 15 for each area. For example, as shown in Fig. 9(a), the shadow direction can be seen as about 30° west of the north of the building. For the same city, we assume that the Sun's position is invariant, and the whole city has the same shadow direction. After that, the shadow mask is directionally expanded according to the calculated angles using the expansion convolution. Therefore, a complementary mask was constructed by extending along a specific direction, which is set according to the shadow formation angle of each region Fig. 9(e).

*Testing Data:* To verify the robustness of the model, six images (see Fig. 10) were selected from each region in the test set. We not only evaluated the network performance qualitatively by visualizing the results but also quantitatively evaluated the results of land-cover classification after shadow removal (see second row in Fig. 10). Furthermore, as shown in the red boxed areas in the first row of Fig. 10, in order to accurately determine the shadow recovery of different features, we selected the same feature categories in shadow and nonshadow areas for texture restoration analysis, which could better validate the generalization capacity of the model for complex features.

### C. Experimental Setup

*1) Evaluation Metrics:* To precisely quantify the degree of shadow recovery, providing a basis for urban observation by pixel, we labeled the land-cover classification maps of six cities and utilized an unsupervised $K$-means clustering method to classify the image after shadow removal. Each predicted image is labeled with three categories: building, grass, and road, and four indicators are used for quantitative evaluation, including OA, and kappa. The formula is shown as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}} \tag{23}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA} - \mathrm{PRE}}{1 - \mathrm{PRE}} \tag{24}$$

$$\mathrm{PRE} = \frac{(\mathrm{TP}+\mathrm{FP}) \times (\mathrm{TP}+\mathrm{FN}) + (\mathrm{TN}+\mathrm{FN}) \times (\mathrm{TN}+\mathrm{FP})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})} \tag{25}$$

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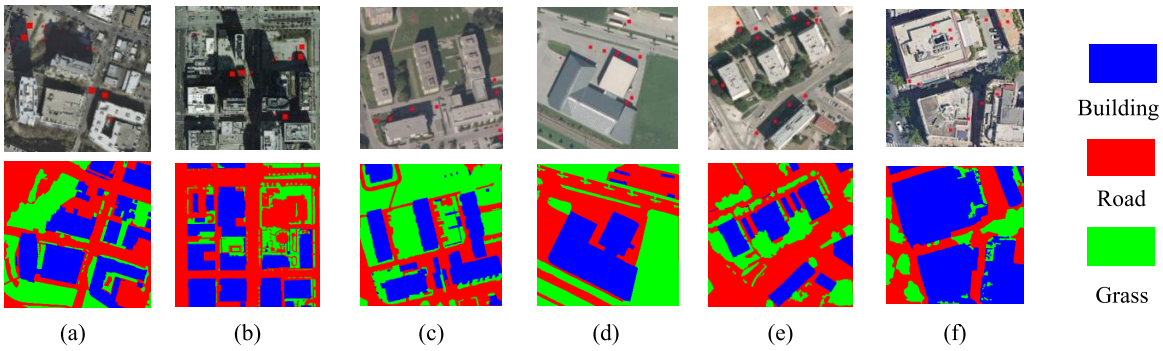


Fig. 10. Manually labeled classification maps. (a) Austin. (b) Tyrol. (c) Vienna. (d) Chicago. (e) Innsbruck. (f) Rome.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \times 100\% \tag{26}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \times 100\% \tag{27}$$

$$F1\text{-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \times 100\% \tag{28}$$

where TP stands for the number of pixels that are accurately classified, TN denotes the number of negative samples that are correctly identified, FN represents the count of positive samples that are misclassified, and FP indicates the number of nontarget samples that are incorrectly classified. Precision is concerned with the proportion of data predicted to be positive (which may contain negative samples), which are true positive samples. Recall concerns the percentage of positive samples (excluding negative samples) correctly predicted in the data.

The $F1$ index considers both the accuracy and the completeness of the check, which allows both to be maximized.

OA means that the ratio of correctly classified pixels in all categories is equal to all pixels. The kappa coefficient considers the consistency of the classification results with the true value.

In addition, the shadow texture recovery (SRI) score is defined after removing shadows by calculating the color difference between similar nonshadow and shadow removal areas [45], which is defined as

$$\gamma = \frac{1}{C}\sum_{c=1}^{C}\frac{1}{N}\left(\sum_{i=1}^{N}\left|\text{Mean}_{c,i}^{\text{Rsd}} - \text{Mean}_{c,i}^{\text{Nsd}}\right|\right) \tag{29}$$

where $C$ represents the channel of the channel image, $N$ represents the number of selected samples, $i$ represents the currently selected pixel, Rsd represents the sample with the shadow removed, and Nsd represents the shadow-free sample. A high SRI score indicates that the shadow-removal regions exhibit significant variance from the nonshadow regions. In contrast, a low SRI score indicates that the shadow-removal regions present high similarity with the nonshadow regions.

*2) Reference Methods:* To verify the superiority of our proposed method, a series of state-of-the-art methods were compared. More specifically, the baselines include not only the traditional methods of the VHR image shadow removal method (HMC) [46] but also the recently proposed method (Color-lines (CL) [47]). In addition, we also reimplement recent shadow removal methods based on weakly supervised deep learning in the natural images (SP + M-Net [34], SP + M + I-Net [23], and G2RShadowNet [48]). To fully evaluate the robustness of our model, we evaluated six cities with different urban structural, including Austin, Vienna, Tyrol, Innsbruck, Chicago, and Rome.

*3) Implementation Details:* In all the generators and discriminators, parameters were initialized using random noise following a zero-mean Gaussian distribution, with a standard deviation set to 0.02. In addition, the network optimization was performed using the Adam optimizer, with the first momentum value set at 0.5 and the second momentum value fixed at 0.999.

The training of the generators spanned 100 epochs, each with a batch size of 1. For the initial 50 epochs, we employed a base learning rate of $2 \times 10^{-4}$, followed by a linear decay reducing the rate to 0 over the remaining epochs. Concurrently, the discriminators underwent training for 100 epochs with a batch size of 1. Here, the base learning rate was set at $8 \times 10^{-4}$ for the first 50 epochs, after which it also underwent a linear decay to 0 for the subsequent epochs.

The training of the network encompasses all components, which interact with each other via forward and backward signal flows. Specifically, during the testing phase, only the shadow preelimination subnet and the prior knowledge-guided refinement subnet are utilized. These subnets work together to generate the final shadow removal result and processing inputs that consist of the shadow image and its corresponding shadow mask. All approaches, including our proposed approach, were implemented by PyTorch and conducted on the single NVIDIA GeForce RTX 2080Ti.

### D. Analysis of the Shadow Removal Experiments

Fig. 11 presents the shadow removal results obtained using various methods. Notably, the SP + M-Net and SP + M + I-Net methods demonstrate an effective approach to shadow elimination. However, these methods yield exceptional results predominantly in regions such as Tyrol and Vienna, which are characterized by simpler class compositions within the shadowed areas. In contrast, when dealing with shadowed scenes encompassing more complex features, these methods do not adequately address the variations among different categories. Consequently, the removal process tends to produce a masking effect, resulting in less distinguishable feature types and still visible shadow areas. The HMC method recovers the hue information of the shadow region based on the histogram of the nonshadow region, which is better than the two previous

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Fig. 11. Shadow removal results' visualization. (a) Image. (b) SP + M-Net. (c) SP + M + I-Net. (d) HMC. (e) G2RShadow. (f) CL. (g) Ours.

methods in recovering the shadow feature. However, due to the disregard of hue consistency between the shadow and nonshadow regions, there exists a large spectral difference in the recovered shadow. Compared with other methods, G2RShadowNet can maintain spectral consistency. However, its results present overall tones and a serious loss of internal feature category information. CL can also ensure the consistency of the nonshadow areas with the nonshadow tones by manually defining the shadow and nonshadow areas. However, the different selected locations will lead to the restoration failure of some feature information, which produces all-black areas. In addition, due to a certain bias in manual labeling, the above method produces a significant edge phenomenon when removing shadows. Compared with the above methods, our proposed method is closer to the nonshadow areas and maintains complete information about the internal feature classes. The most significant reason is attributed to the prior knowledge of the nonshadow areas in specific directions and the effective combination of feature preservation and spectral consistency information. Besides, the more natural transition between the shadow edges and the nonshadow areas is due to the use of mean filtering for edge correction.

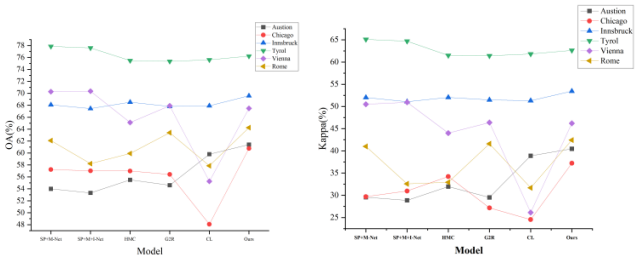


Fig. 12. Land-cover classification results in six cities. (a) Impact on OA of different cities. (b) Impact on kappa of different cities.

### *E. Quantitative Comparison of the Different Models*

To quantify the shadow removal effect more intuitively, we use an unsupervised $K$-means clustering method to evaluate the land-cover classification maps. As shown in Table I, the classification accuracy of our method achieves higher OA and kappa, which demonstrates that the incorporation of prior knowledge enables the simultaneous consideration of both texture and spectral features.

Fig. 12 shows a more intuitive classification of each region. Compared with other methods, both the OA and kappa

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TABLE I
EVALUATION OF AVERAGE LAND-COVER CLASSIFICATION RESULT IN SIX CITIES

| | | SP+M-Net[34] | SP+M+I-Net[23] | HMC[46] | G2RShadowNet[48] | CL[47] | **Ours** |
|---|---|---|---|---|---|---|---|
| Land Cover | OA | 64.92 | 63.99 | 63.58 | 64.24 | 60.75 | **66.62** |
| | Kappa | 44.64 | 43.18 | 42.77 | 42.92 | 39.06 | **47.06** |
| Building | F1 | 69.71 | 65.53 | 77.71 | 76.16 | **81.91** | 81.02 |
| Road | F1 | 81.40 | 80.58 | 83.29 | 82.03 | 82.94 | **86.02** |

coefficients of our proposed method remain superior and stable in six cities, which proves the excellence of our method.

Furthermore, specialized networks are employed for the extraction of specific classes of images following shadow removal. These include the multiscale-aware and segmentation-prior conditional random fields (MSCRFs) for advanced building extraction [49] and the global context-aware and batch-independent network (GCBNet) for road extraction [50]. Each of these networks is tailored to their respective tasks, enhancing the precision and effectiveness of the image extraction process. From the table, our method achieves $F1$ scores of 81.02 and 86.02, respectively. Compared to other methods, our method recovers the color and texture of each category better, making it closer to the real unshaded situation. For categories such as roads that frequently encounter extensive shadows, the comparative accuracy data indicate a significant advantage of our method. It substantially improves the model's ability to pinpoint road locations. In contrast, alternative methods often fail to detect roads in shadowed areas, a limitation primarily attributed to their inadequate shadow recovery capabilities.

Still, as shown in Fig. 11 (Vienna), the internal detail texture retention is not as excellent as ours, and the KO-Shadow can better guarantee the diversity of feature classes with consistent color. In general, our method achieves more robust removal results for areas with different architectural styles and distribution of land cover.

### *F. Generalizability Analysis for Complex Scenario Mapping*

As shown in Fig. 13, a large image containing more complex categories was used to verify the generalization of the proposed method. The size of Fig. 13(a) is 4096 × 2048, and the corresponding mask has been labeled in detail. Compared with the result obtained by the computer vision method (G2RShadowNet), the proposed KO-Shadow has lower chromatic aberration. Specifically, the amplified results in Fig. 13(d) left show that KO-Shadow obtains closer to the true spectral information of nonshadow areas by progressive removal and supplementary information on tones in nonshadow areas. In contrast, the computer vision method will have a haze over the shadow areas. As can be seen in Fig. 13(d) right, the amplified result shows that the KO-Shadow recovers the hue of the shadow areas while ensuring that other feature categories remain consistent. The KO-Shadow represents an effective solution for large-scale VHR image shadow removal by overcoming the imaging differences in texture and spectra of VHR images.

### *G. Texture Restoration and Complexity Analysis*

To verify that the effect of various methods on the color of the removed features is consistent with those of the

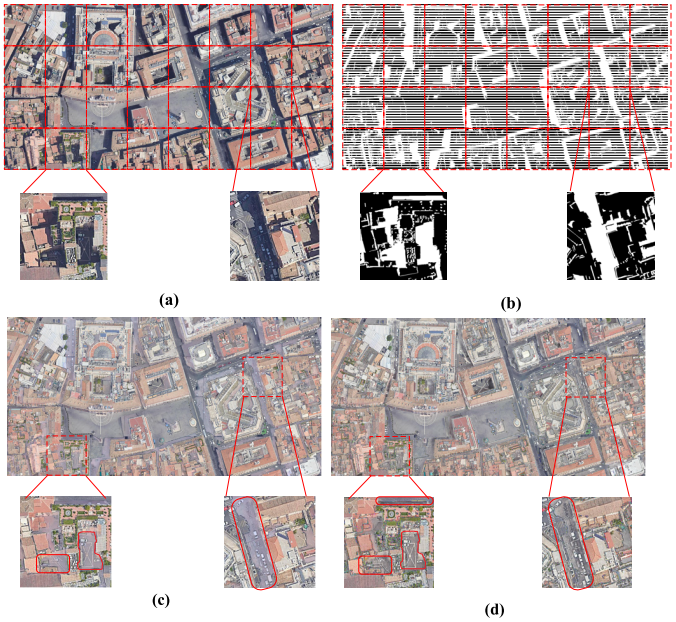


Fig. 13. Visualizations of the removal results for complex scenario mapping. (a) Image. (b) Mask. (c) Computer vision method (G2RShadowNet). (d) Ours.

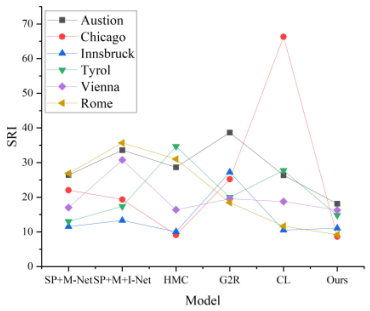


Fig. 14. Comparison of SRI scores of all methods in different cities.

shadow-free areas, a new score (SRI) has been developed to calculate the difference in each feature type.

Due to the lack of classification labels, it is unable to compare all the features of the same category located in the shade removal region and the true nonshadow region. As shown in Fig. 14, the SRI score was compared based on the manual selection of purely similar species. The matched samples of feature types in different regional images were randomly selected, and as shown in Table II, our method achieved 18.125, 8.629, 11.06, 14.80, 16.28, and 9.21 SRI scores in Austin, Chicago, Innsbruck, Tyrol, Vienna, and Rome, respectively.

As shown in Fig. 14, our method achieves a small SRI score in most cities, which indicates that the similarity of our method after removing shadows, and the feature retention is better and closer to the real situation. The main contribution is

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TABLE II
COMPARISON OF SRI SCORE AND COMPLEXITY OF DIFFERENT CITIES USING DIFFERENT METHODS

| Area | Austin | Chicago | Innsbruck | Tyrol | Vienna | Rome | Complexity(s) |
|---|---|---|---|---|---|---|---|
| SP+M-Net[34] | 26.4 | 22.06 | 11.53 | **13.00** | 17.05 | 26.89 | 0.1970 |
| SP+M+I-Net[23] | 33.6 | 19.33 | 13.33 | 17.33 | 30.78 | 35.67 | 0.2169 |
| HMC[46] | 28.66 | 9.11 | **10.06** | 34.67 | 16.38 | 31.00 | 3.0912 |
| G2RShadowNet[48] | 38.66 | 25.22 | 27.26 | 20.00 | 19.57 | 18.45 | 0.4498 |
| CL[47] | 26.33 | 66.33 | 10.53 | 27.73 | 18.76 | 11.67 | 3.2589 |
| Ours | **18.12** | **8.62** | 11.06 | 14.80 | **16.28** | **9.21** | **0.1607** |

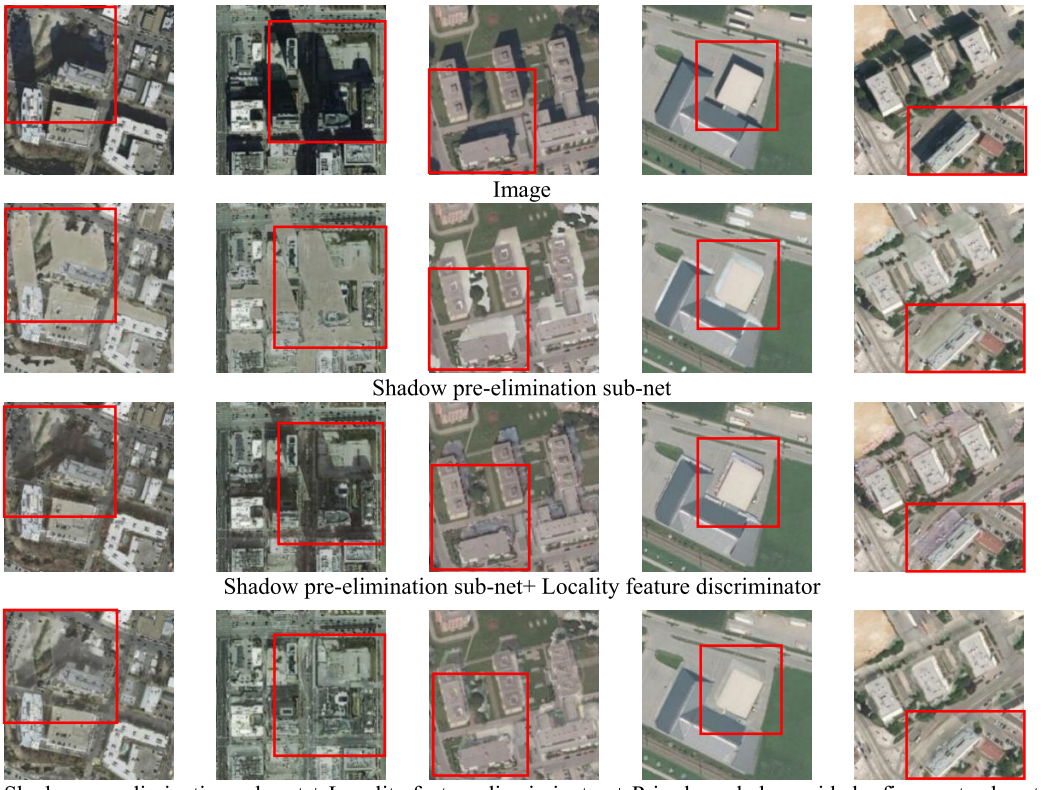


Fig. 15. Ablation experimental results. (a) Austin. (b) Chicago. (c) Innsbruck. (d) Tyrol. (e) Vienna.

that the color consistency module can ensure the consistency of different regions, making the recovered region more similar to other nonshadowed regions.

In order to validate the complexity of models, we calculated the runtime of a single image for each method. As shown in Table II, the traditional methods (HMC and CL) have a long processing speed due to the complex computational process. For deep learning methods, our methods have a shorter runtime for testing due to only using trained $G_s$ and $G_f$ for testing and having a smaller number of model parameters.

### H. Ablation Experiment

To further analyze the mechanism of our method, the ablation study was conducted to evaluate the specific improvement of the key components. In Table III, *A* represents the preelimination subnet, *B* represents the locality feature discriminator, and *C* represents the shadow refinement subnet. As shown in Fig. 15 and Table III, the shadow preelimination subnet simply brightens the shadow areas to make the shadow and nonshadow areas uniform in color and is difficult to match the reality and distinguish between features within the shadow areas, with SRI value reflecting a large discrepancy. With the addition of a locality feature discriminator, the model has a better knowledge of shadows and nonshadows, which makes the brightness of the image closer to the real nonshadowed region, and its SRI value also has a certain improvement. Finally, with the addition of a prior knowledge-guided refinement subnet, the model can better recover the spectra and

TABLE III
COMPARISON OF SRI VALUES AFTER REMOVAL OF SHADOWS FOR DIFFERENT COMPONENTS

| A | B | C | Austin | Chicago | Innsbruck | Tyrol | Vienna |
|---|---|---|---|---|---|---|---|
| √ | | | 47.58 | 25.29 | 52.73 | 43.4 | 27.09 |
| √ | √ | | 30.54 | 15.82 | 26.4 | 21.13 | 25.28 |
| √ | √ | √ | **18.125** | **8.629** | **11.06** | **14.80** | **16.28** |

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features of different categories within the shadowed region according to the real nonshadowed region, and the SRI value also achieves the highest value.

## V. Conclusion

This article proposed a shadow removal framework for VHR imagery urban areas via weakly supervised named KnOwledge-driven shadow progressive removal framework (KO-Shadow). In KO-Shadow, a shadow preelimination subnet is used to initially eliminate shadows, and then, the spectral and texture information of the shadow regions is optimized more carefully by a prior knowledge-guided refinement subnet. Finally, the operation of the two generators is supervised by a locality feature discriminator, which performs feature extraction on the mask-covered area. Experimental results demonstrate that KO-Shadow is more robust and effective. Furthermore, a generalizability analysis is carried out on complex scenes to verify the model's capacity to maintain texture information after shadow removal. Remarkably, KO-Shadow exhibits a more tailored restoration effect for different surface types. However, when the colors of the shadow objects are closer to each other, our proposed method still fails to restore their original state but uniformly restores them to a single color. In future work, we intend to distinguish similar colors more finely and restore them. Finally, we intend to integrate shadow detection and shadow removal in an end-to-end framework for VHR images.

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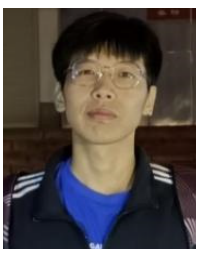

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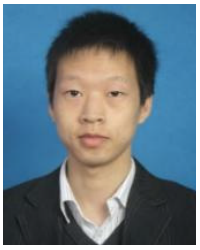

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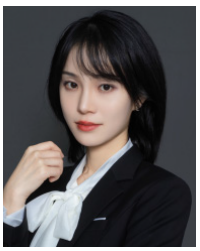

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